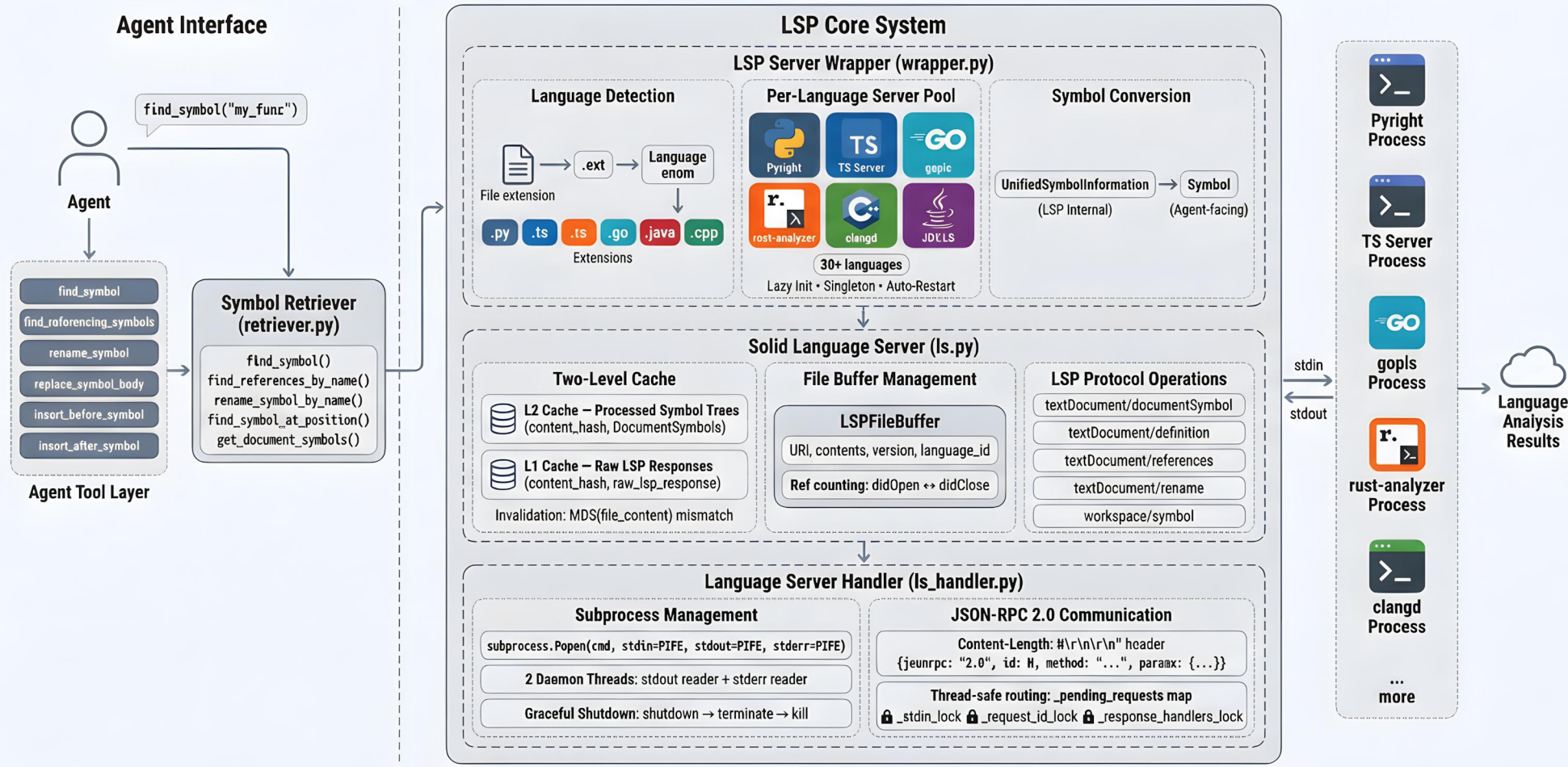
Agent Interface
find_symbol("my_func")
Agent
find_symbol
find_roforencing_symbols
rename_symbol
replace_symbol_body
insort_before_symbol
insort_after_symbol
Agent Tool Layer
Symbol Retriever (retriever.py)
find_symbol()
find_references_by_name()
rename_symbol_by_name()
find_symbol_at_position()
get_document_symbols()
LSP Core System
LSP Server Wrapper (wrapper.py)
Language Detection
File extension
.ext
Language enom
.py
.ts
.ts
.go
.java
.cpp
Extensions
Per-Language Server Pool
Pyright
TS Server
gopic
rost-analyzer
clangd
JDCLS
30+ languages
Lazy Init • Singleton • Auto-Restart
Symbol Conversion
UnifiedSymbolInformation
(LSP Internal)
Symbol
(Agent-facing)
Solid Language Server (ls.py)
Two-Level Cache
L2 Cache — Processed Symbol Traes (content_hash, DocumentSymbols)
L1 Cache — Raw LSP Responses (content_hash, raw_lsp_response)
Invalidation: MDS(file_content) mismatch
File Buffer Management
LSPFileBuffer
URI, contents, version, language_id
Ref counting: didOpen ↔ didClose
LSP Protocol Operations
textDocument/documentSymbol
textDocument/definition
textDocument/references
textDocument/rename
workspace/symbol
Language Server Handler (ls_handler.py)
Subprocess Management
subprocess.Popen(cmd, stdin=PIFE, stdout=PIFE, stderr=PIFE)
2 Daemon Threads: stdout reader + stderr reader
Graceful Shutdown: shutdown → terminate → kill
JSON-RPC 2.0 Communication
Content-Length: #\r\n\r\n" header
{jeunrpc: "2.0", id: H, method: "...", paraax: {...}}
Thread-safe routing: _pending_requests map
_stdin_lock _request_id_lock _response_handlers_lock
stdin
stdout
Pyright Process
TS Server Process
gopls Process
rust-analyzer Process
clangd Process
...
more
Language Analysis Results